

Navier-Stokes equations

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July 7, 2026

Goal

This project solves, by means of the finite element method, the unsteady, incompressible Navier-Stokes equations to simulate the benchmark problem "flow past a cylinder" [1] for different values of the Reynolds number $\text{Re} \leq 200$, together with the computation of the drag and lift coefficients acting on the obstacle over time.

1 Introduction

We formulate the weak form of the problem and its discretization for a generic spatial dimension $d \in \{2, 3\}$, since the same mathematical model and numerical scheme are used for both the two-dimensional and the three-dimensional benchmark configurations.

We consider a domain $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, with boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$, $\Gamma_D \cap \Gamma_N = \emptyset$. The problem consists in finding the velocity field $\mathbf{u} : \Omega \times (0, T] \rightarrow \mathbb{R}^d$ and the pressure field $p : \Omega \times (0, T] \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mathbf{f} & \text{in } \Omega, \\ \nabla \cdot \mathbf{u} = 0 & \text{in } \Omega, \\ \mathbf{u} = \mathbf{g} & \text{on } \Gamma_D, \\ \nu \nabla \mathbf{u} \mathbf{n} - p \mathbf{n} = \mathbf{h} & \text{on } \Gamma_N, \\ \mathbf{u}(t=0) = \mathbf{u}_0 & \text{in } \Omega, \end{cases} \quad (1)$$

where ν is the kinematic viscosity, $\mathbf{f}$ a forcing term for unit of mass (taken as zero in the cylinder benchmark), and $\mathbf{g}$ the prescribed Dirichlet velocity: a parabolic inflow profile at the channel inlet and a no-slip condition on the channel walls and on the surface of the cylindrical obstacle. On the outflow boundary Γ_N a homogeneous Neumann condition $\mathbf{h} = \mathbf{0}$ is imposed, as it is a standard choice for this benchmark [1]. The same formulation applies whether Ω is the two-dimensional channel cross-section or the full three-dimensional channel with a cylindrical obstacle, the only differences being the geometry, the mesh, and the definitions of the drag and lift coefficients (*see below*).

The system (1) is a time-dependent, nonlinear, saddle-point problem. Both the latter two features

directly shape the choices made in the remainder of this section: the convective nonlinearity motivates the *semi-implicit* time-linearization of Section 1.3, while the saddle-point structure motivates the compatibility (inf-sup) requirement on the finite element spaces discussed in Section 1.2.

1.1 Weak formulation

To state the weak formulation precisely, we first introduce the relevant function spaces. Given

$$H^1(\Omega) = \left\{ v \in L^2(\Omega) : \nabla v \in [L^2(\Omega)]^d \right\},$$

we define the space of vector-valued test functions vanishing on the Dirichlet boundary,

$$\mathbf{V}_0 = \left\{ \mathbf{v} \in [H^1(\Omega)]^d : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D \right\}.$$

The velocity solution $\mathbf{u}$, however, must satisfy the Dirichlet condition $\mathbf{u} = \mathbf{g}$ on Γ_D : we therefore introduce the affine space

$$\mathbf{V}_{\mathbf{g}} = \left\{ \mathbf{v} \in [H^1(\Omega)]^d : \mathbf{v} = \mathbf{g} \text{ on } \Gamma_D \right\}.$$

Finally, for the pressure we take $Q = L^2(\Omega)$, with no boundary conditions imposed since the pressure only appears through its gradient in the momentum equation.

It is convenient to introduce explicitly the bilinear and trilinear forms that will recur throughout the discretization. For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in [H^1(\Omega)]^d$ and $p, q \in L^2(\Omega)$, we define

$$a(\mathbf{u}, \mathbf{v}) := \nu \int_{\Omega} \nabla \mathbf{u} : \nabla \mathbf{v} \, d\Omega, \quad (2)$$

$$b(\mathbf{v}, q) := - \int_{\Omega} q (\nabla \cdot \mathbf{v}) \, d\Omega, \quad (3)$$

$$c(\mathbf{w}; \mathbf{u}, \mathbf{v}) := \int_{\Omega} [(\mathbf{w} \cdot \nabla) \mathbf{u}] \cdot \mathbf{v} \, d\Omega. \quad (4)$$

Here $a(\cdot, \cdot)$ is the (bilinear, symmetric, coercive on $\mathbf{V}_0$) viscous form, $b(\cdot, \cdot)$ is the (bilinear) pressure-velocity coupling form responsible for the saddle-point structure, and $c(\mathbf{w}; \cdot, \cdot)$ is the (trilinear, linear in each of

its three arguments, but nonlinear overall once $\mathbf{w} = \mathbf{u}$ convective form. With this notation, the weak formulation of (1) reads: for each $t \in (0, T]$, find $(\mathbf{u}(t), p(t)) \in \mathbf{V}_{\mathbf{g}} \times Q$ such that for all $(\mathbf{v}, q) \in \mathbf{V}_0 \times Q$

$$\begin{cases} \left(\frac{\partial \mathbf{u}}{\partial t}, \mathbf{v} \right) + a(\mathbf{u}, \mathbf{v}) + b(\mathbf{v}, p) + c(\mathbf{u}; \mathbf{u}, \mathbf{v}) \\ \qquad \qquad \qquad = (\mathbf{f}, \mathbf{v}) + (\mathbf{h}, \mathbf{v})_{\Gamma_N} \\ b(\mathbf{u}, q) = 0 \end{cases} \quad (5)$$

which is exactly the form obtained by multiplying the momentum equation by a test velocity $\mathbf{v}$ and the continuity equation by a test pressure q , integrating over Ω , and integrating by parts the viscous and the pressure terms. The Dirichlet condition on $\mathbf{u}$ is enforced strongly in the definition of $\mathbf{V}_{\mathbf{g}}$, and the Neumann condition enters naturally as a boundary integral on the right-hand side. The solution $\mathbf{u} \in \mathbf{V}_{\mathbf{g}}$ carries indeed the physical boundary data, while test functions $\mathbf{v} \in \mathbf{V}_0$ are homogeneous, so that the boundary terms arising from integration by parts reduce to the prescribed Neumann datum on Γ_N alone.

1.2 Spatial discretization

At the discrete level, this translates directly. As a matter of fact, let us introduce the finite elements space

$$X_h^r = \{\mathbf{v}_h \in \mathcal{C}^0(\bar{\Omega}) \mid \mathbf{v}_h|_K \in \mathbb{P}^r, \forall K \in \mathcal{T}_h\},$$

where $\mathbb{P}^r$ is the space of polynomials of degree r , and $\mathcal{T}_h$ is a regular triangulation approximating Ω . We can therefore define

$$\begin{aligned} \mathbf{V}_h &= (X_h^2)^d \cap \mathbf{V} \\ Q_h &= X_h^1 \cap Q, \end{aligned}$$

being respectively the finite-dimensional spaces for velocity and pressure.

The domain is discretized with a mesh $\mathcal{T}_h$ of simplices (triangles in 2D, tetrahedra in 3D), and the velocity–pressure pair is approximated with an *inf-sup stable* Taylor–Hood-type element: continuous piecewise polynomials of degree $k+1$ for each velocity component and degree k for the pressure, i.e. $\mathbf{V}_h \times Q_h = [P_{k+1}]^d \times P_k$, both built on the same mesh $\mathcal{T}_h$.

In particular, because of the saddle-point structure highlighted above, an *equal-order* pairing is in general *not* numerically stable, and can produce spurious pressure oscillations (the discrete analogue of the pressure not being uniquely determined). Stability and well-posedness of the discrete saddle-point problem require indeed the pair $(\mathbf{V}_h, Q_h)$ to satisfy the discrete inf-sup (Ladyzhenskaya–Babuška–Brezzi, LBB) condition

$$\inf_{q_h \in Q_h \setminus \{0\}} \sup_{\mathbf{v}_h \in \mathbf{V}_h \setminus \{0\}} \frac{b(\mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1} \|q_h\|_{L^2}} \geq \beta_h > 0, \quad (6)$$

with β_h bounded away from zero uniformly when $h \rightarrow 0$. The Taylor–Hood pair $[P_{k+1}]^d \times P_k$ ($k \geq 1$) is a classical example satisfying (6), and yielding the following error estimate

$$\begin{aligned} \|\mathbf{u} - \mathbf{u}_h\|_{[H^1(\Omega)]^d} + \|p - p_h\|_{L^2(\Omega)} \\ \leq Ch^2(|\mathbf{u}|_{[H^3(\Omega)]^d} + |p|_{[H^2(\Omega)]^d}). \end{aligned}$$

The semi-discrete Galerkin problem then reads: for each $t \in (0, T]$, find $(\mathbf{u}_h(t), p_h(t)) \in \mathbf{V}_{\mathbf{g},h} \times Q_h$ such that for all $(\mathbf{v}_h, q_h) \in \mathbf{V}_{0,h} \times Q_h$

$$\begin{cases} \left(\frac{\partial \mathbf{u}_h}{\partial t}, \mathbf{v}_h \right) + a(\mathbf{u}_h, \mathbf{v}_h) + c(\mathbf{u}_h; \mathbf{u}_h, \mathbf{v}_h) + \\ \qquad \qquad \qquad + b(\mathbf{v}_h, p_h) = (\mathbf{f}, \mathbf{v}_h) + (\mathbf{h}, \mathbf{v}_h)_{\Gamma_N} \\ b(\mathbf{u}_h, q_h) = 0 \end{cases} \quad (7)$$

Quadrature (Gauss rules on simplices) is used for both cell and face integrals, with the number of quadrature points scaled to the polynomial degree in order to integrate the nonlinear convective term $c(\mathbf{u}_h; \mathbf{u}_h, \mathbf{v}_h)$ accurately.

1.3 Time discretization

Time is discretized with the generalized θ -method. Denoting by $\mathbf{u}^n \approx \mathbf{u}(t_n)$ the discrete velocity at time $t_n = n \Delta t$ for $n = 0, \dots, N_T$ with N_T number of sub-intervals of $[0, T]$, the time derivative is approximated using forward finite differences and each term of the Galerkin formulation is treated implicitly with weight θ and explicitly with weight $(1 - \theta)$. The fully-discrete problem finally reads: find $(\mathbf{u}_h^{n+1}, p_h^{n+1}) \in \mathbf{V}_{\mathbf{g},h} \times Q_h$ such that for all $(\mathbf{v}_h, q_h) \in \mathbf{V}_{0,h} \times Q_h$

$$\begin{cases} \left(\frac{\mathbf{u}_h^{n+1} - \mathbf{u}_h^n}{\Delta t}, \mathbf{v}_h \right) + \\ \theta [a(\mathbf{u}_h^{n+1}, \mathbf{v}_h) + c(\mathbf{u}_h^{n+1}; \mathbf{u}_h^{n+1}, \mathbf{v}_h)] + \\ (1 - \theta) [a(\mathbf{u}_h^n, \mathbf{v}_h) + c(\mathbf{u}_h^n; \mathbf{u}_h^n, \mathbf{v}_h)] + \\ b(\mathbf{v}_h, p_h^{n+1}) = \\ \theta (\mathbf{f}^{n+1}, \mathbf{v}_h) + (1 - \theta) (\mathbf{f}^n, \mathbf{v}_h) + \\ (\mathbf{h}, \mathbf{v}_h)_{\Gamma_N} \\ b(\mathbf{u}_h^{n+1}, q_h) = 0 \end{cases}, \quad (8)$$

recovering the *unconditionally stable* backward Euler scheme for $\theta = 1$. The pressure gradient (and, correspondingly, the incompressibility constraint) is always treated fully implicitly at t_{n+1} , regardless of θ : since the pressure carries no time derivative of its own, there is no explicit counterpart to blend, and enforcing $\nabla \cdot \mathbf{u}_h^{n+1} = 0$ is what guarantees that the computed velocity field is divergence-free at the *new* time level, consistently with the implicit treatment of the saddle-point coupling in Eq. (10) below.

To avoid solving a nonlinear system at every time step, the convective term is linearized *semi-implicitly*,

following the standard approach also adopted in [2]: the transport velocity is taken from the previous time step, $(\mathbf{u}_h^n \cdot \nabla) \mathbf{u}_h^{n+1}$ for the implicit part and $(\mathbf{u}_h^n \cdot \nabla) \mathbf{u}_h^n$ for the explicit part, again blended with weight θ , i.e. the nonlinear term $c(\mathbf{u}_h^{n+1}; \mathbf{u}_h^{n+1}, \mathbf{v}_h)$ of the discrete problem (8) is replaced at each time step by

$$\theta c(\mathbf{u}_h^n; \mathbf{u}_h^{n+1}, \mathbf{v}_h) + (1 - \theta) c(\mathbf{u}_h^n; \mathbf{u}_h^n, \mathbf{v}_h), \quad (9)$$

which is now *linear* in the unknown $\mathbf{u}_h^{n+1}$.

However, substituting $\mathbf{u}_h^n = \mathbf{u}_h^{n+1} + O(\Delta t)$ into (9) shows that the linearized form differs from the exact nonlinear term $c(\mathbf{u}_h^{n+1}; \mathbf{u}_h^{n+1}, \mathbf{v}_h)$ by a residual of order $O(\Delta t)$, so the overall time-stepping scheme is limited to first-order accuracy in time once the semi-implicit treatment of convection is taken into account. This is a deliberate trade-off in exchange for linearity at each step, and is a standard practice for moderate Reynolds number flows such as the present benchmark ($\text{Re} \leq 200$), where the resulting time-discretization error remains controlled provided Δt is chosen sufficiently small.

1.4 Algebraic system and solver

Expanding (8) on a finite element basis produces, at each time step, a block linear system of saddle-point type

$$\begin{pmatrix} F & B^T \\ B & 0 \end{pmatrix} \begin{pmatrix} \mathbf{U}_h^{n+1} \\ P_h^{n+1} \end{pmatrix} = \begin{pmatrix} G \\ 0 \end{pmatrix}, \quad (10)$$

where the $F = \frac{1}{\Delta t} M_u + \theta[A_u + C(\mathbf{u}^n)]$ collects the mass, viscous, and linearized convective contributions on the velocity block, B (and its transpose) the pressure-velocity coupling from the divergence/gradient terms, and G is a known vector depending on the discretized source force $\mathbf{f}$, on $\mathbf{U}_h^n$, and on Neumann boundary data. Dirichlet conditions on the inflow, walls, and cylinder are indeed imposed strongly on the velocity block, while the homogeneous Neumann condition at the outflow contributes to the right-hand side. The resulting non-symmetric indefinite system is solved with a Krylov method (FGMRES) preconditioned by aSIMPLE preconditioner.

The inf-sup condition (6) satisfied by the Taylor-Hood pair is precisely what guarantees that the block matrix in (10) is nonsingular; without an inf-sup stable pair, the algebraic system could become numerically singular or severely ill-conditioned as the mesh is refined, independently of the linear solver used.

1.5 aSIMPLE preconditioner

The Semi-Implicit Method for Pressure Linked Equations (SIMPLE) has been introduced as an iterative method which first solves the momentum equation and then updates the pressure and the velocity field to conserve the mass by using the continuity equation. This

method is reinterpreted by [2] as if it were associated to a preconditioner

$$P_{\text{SIMPLE}} = \begin{pmatrix} F & 0 \\ B & -\tilde{S} \end{pmatrix} \begin{pmatrix} I & D^{-1}B^T \\ 0 & \alpha I \end{pmatrix}, \quad (11)$$

where D is the diagonal of F , $\alpha \in (0, 1]$ is a parameter that damps the pressure update, and $\tilde{S} = BD^{-1}B^T$.

In order to find a balance between keeping the number of iterations as low as possible, while trying to decrease the wall time as much as possible, it may be more efficient to apply a less accurate preconditioner. To do so, the approximate SIMPLE (aSIMPLE) preconditioner introduced in [2] replaces inverses of the algebraic operators with some suitable approximations denoted by a *hat*:

$$P_{\text{aSIMPLE}}^{-1} = \begin{pmatrix} D^{-1} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -B^T \\ 0 & I \end{pmatrix} \begin{pmatrix} D & 0 \\ 0 & \frac{1}{\alpha} I \end{pmatrix} \times \begin{pmatrix} I & 0 \\ 0 & -\hat{\tilde{S}}^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -B & I \end{pmatrix} \begin{pmatrix} \hat{F}^{-1} & 0 \\ 0 & I \end{pmatrix}. \quad (12)$$

1.6 Benchmark quantities

Following the definitions of [1], the quality of the numerical solution is assessed through the drag and lift coefficients

$$c_D = \frac{F_D}{\bar{U}^2 D} \quad c_L = \frac{F_L}{\bar{U}^2 D}, \quad (13)$$

where F_D and F_L are the forces exerted by the fluid on the cylinder in the streamwise and cross-stream directions respectively, $\bar{U}$ is a reference velocity, and D is the cylinder diameter. In particular, the drag F_D is a resistive force opposing the motion of the body through the fluid, while the lift force F_L is responsible for pushing it perpendicular to the flow.

For a symmetric body perfectly aligned with the oncoming fluid flow, the geometry presented to the fluid is an exact mirror image above and below the centerline. This forces the resulting fluid flow field and streamlines to be perfectly symmetric as well. Consequently, the fluid creates identical velocity distributions on both sides and, by extension, identical pressure distributions on the upper and lower surfaces. Since lift is the net perpendicular force obtained by integrating these pressures over the body's surface, the downward force exerted on the top is exactly countered by the upward force on the bottom. Since these vertical forces cancel each other out completely at every corresponding point, the total lift force is zero.

Depending on the Reynolds number, the flow is either steady (low Re , benchmark case 2D-1/3D-1Z) or exhibits periodic vortex shedding (higher Re , benchmark cases 2D-2/2D-3 and 3D-2Z/3D-3Z), in which case additional quantities such as the Strouhal number St , the maximum drag/lift coefficients, and the pressure difference across the cylinder at a prescribed phase

of the shedding cycle are also computed and compared against the reference intervals reported in [1].

1.7 Reynolds number

The Reynolds number characterizes the ratio of inertial to viscous forces in the flow and is defined, for both the 2D and 3D benchmark configurations, as

$$\text{Re} = \frac{\bar{U}D}{\nu}, \quad (14)$$

where D is the characteristic length of the obstacle (the cylinder diameter), ν is the kinematic viscosity of the fluid, and $\bar{U}$ is a reference velocity. Following [1], in 2D the latter is defined from the centerline inflow velocity as

$$\bar{U}(t) = \frac{2}{3} U(0, H/2, t), \quad (15)$$

while in 3D it is defined analogously from the value at the center of the inflow cross-section

$$\bar{U}(t) = \frac{4}{9} U(0, H/2, H/2, t), \quad (16)$$

with H the channel height. For the steady test cases, the inflow (and hence $\bar{U}$) is constant in time, so Re in (14) is a single fixed number; for the unsteady test cases with a time-modulated inflow, $\bar{U}$ and consequently Re vary with t , and the reported value of Re refers to the reference (peak) inflow amplitude used to define the benchmark case.

1.8 Strouhal number

The Strouhal number is a dimensionless quantity that characterizes the frequency of vortex shedding behind the obstacle once the Reynolds number exceeds the threshold for the onset of unsteadiness (test cases 2D-2/2D-3 and 3D-2Z/3D-3Z), relating the shedding frequency f to the characteristic length D and reference velocity $\bar{U}$ of the flow via

$$\text{St} = \frac{fD}{\bar{U}}. \quad (17)$$

In practice, f is not known a priori and must be extracted a posteriori from the computed time history of the lift coefficient $c_L(t)$, which oscillates periodically once the transient has decayed and a stable shedding pattern has been reached. Denoting by $\{t_k\}_{k=1}^M$ the sequence of time instants at which $c_L(t)$ attains a local maximum above a suitable threshold (after discarding an initial transient window), the mean shedding period is estimated as the average spacing between consecutive peaks,

$$T = \frac{1}{M-1} \sum_{k=1}^{M-1} (t_{k+1} - t_k), \quad (18)$$

and the shedding frequency and Strouhal number follow as

$$f = \frac{1}{T}, \quad \text{St} = \frac{Df}{\bar{U}} = \frac{D}{T\bar{U}}. \quad (19)$$

This peak-detection approach requires at least two well-separated maxima of $c_L(t)$ to be resolved within the simulated time window; if fewer than two peaks are detected (e.g., because the flow has not yet reached a periodic regime, or the Reynolds number is too low for shedding to occur), St is undefined.

2 Implementation

The solver is implemented in `deal.II` and organized into two dimension-specific classes, `NavierStokes2D` and `NavierStokes3D`, which share the same overall structure and differ only in the geometry-dependent parts (mesh, reference-velocity definition, and force normalization).

2.1 Mesh generation

The computational domain and mesh are generated with Gmsh from a `.geo` template and imported via `GridIn::read_msh`. In 2D, the domain is the channel $[0, 2.2] \times [0, 0.41]$ with a circular obstacle of radius 0.05 centered at $(0.2, 0.2)$, discretized with triangles.

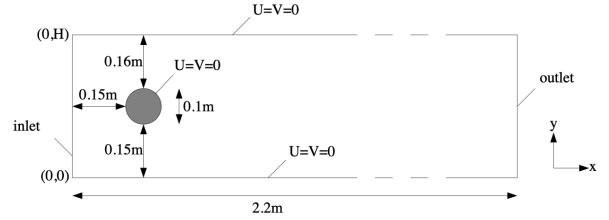


Figure 1: Geometry of 2D test cases with BCs.

In 3D, the same cross-section is extruded along z over a depth $W = 0.41$ to give the domain $[0, 2.5] \times [0, 0.41] \times [0, 0.41]$, discretized with tetrahedra using the Gmsh HXT algorithm.

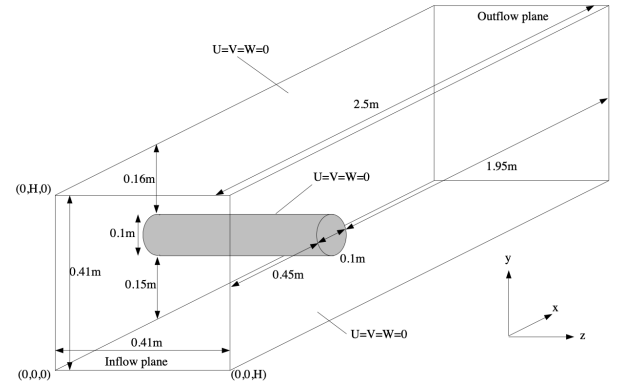


Figure 2: Geometry of 3D test cases with BCs.

In both cases two characteristic mesh sizes are used: a coarser size h_{far} away from the obstacle and a finer

size $h_{\text{circle}} = h_{\text{far}}/2$ on the cylinder boundary, to better resolve the boundary layer and the wake.

Physical boundary tags are assigned during mesh generation and reused directly as `boundary_id` values in the solver:

ID	2D	ID	3D
1	Inlet	1	Inflow
2	Outlet	2	Outflow
3	Top/bottom walls	3	Walls
4	Cylinder	4	Cylinder

2.2 Assembly

At each time step, `assemble()` loops over the locally owned cells and, for each quadrature point, adds the contributions corresponding term-by-term to the θ -method weak form of Eq. (8). A face loop over boundary faces with `boundary_id` 2 (outlet) adds the Neumann contribution $(\mathbf{h}, \mathbf{v})_{\Gamma_N}$; in the cylinder benchmark, $\mathbf{h} = \mathbf{0}$, so this reduces to the standard homogeneous outflow condition.

2.3 Boundary conditions

Dirichlet data are imposed strongly, after assembly, via `VectorTools::interpolate_boundary_values` restricted to the velocity components only (the pressure component is excluded from the constraint mask). A parabolic inflow profile `InletVelocity` is imposed on `boundary_id` 1, and a homogeneous (no-slip) condition is imposed both on the channel walls (`boundary_id` 3) and on the cylinder surface (`boundary_id` 4). The constrained values are then eliminated from the linear system through `MatrixTools::apply_boundary_values`.

2.4 Linear solver

The assembled saddle-point system (10) is solved at every time step with the flexible GMRES method (`SolverFGMRES`), which minimizes the residual over a non-symmetric, indefinite Krylov space while allowing the preconditioner to change from one iteration to the next (this property has been exploited when looking for the best preconditioner).

The chosen preconditioner is `aSIMPLE`, described in detail in Section 1.5, and the stopping criterion is a relative tolerance of 10^{-4} on the residual, without restart.

2.5 Time-stepping and output

The `run()` method initializes the flow at rest ($\mathbf{u}_0 = \mathbf{0}$), then advances the solution with a fixed time step Δt until the final time T , calling `assemble()` and `solve()` at each iteration and updating the ghosted solution vector for the next step's evaluation of $\mathbf{u}^n$. The solution is exported to VTU format every 10 time steps

via `DataOut::write_vtu_with_pvtu_record`, for visualization of velocity and pressure fields.

2.6 Benchmark quantities computation

At every time step, the drag and lift forces are computed by integrating the fluid traction over the cylinder surface (`boundary_id` 4). The two implementations differ slightly in how the traction is expressed:

- in `NavierStokes2D`, the tangential-derivative form of [1] is used directly,

$$F_D = \int_S (\rho \nu \partial_n v_t n_y - p n_x) dS, \quad (20)$$

$$F_L = - \int_S (\rho \nu \partial_n v_t n_x + p n_y) dS. \quad (21)$$

with the tangent $\mathbf{t} = (n_y, -n_x)$ and $\partial_n v_t$ computed from the full velocity gradient at each face quadrature point;

- in `NavierStokes3D`, the equivalent (and more general) stress-tensor form is used

$$\begin{aligned} \mathbf{F} &= \int_S \boldsymbol{\sigma} \cdot \mathbf{n} \\ &= \int_S [-p \mathbf{n} + \rho \nu (\nabla \mathbf{u} + \nabla \mathbf{u}^T) \mathbf{n}] dS, \end{aligned} \quad (22)$$

where $\mathbf{n} = (n_x, n_y)$ is the outward normal vector to the surface S , and the drag and lift are the x - and y -components of the integrated traction.

The Reynolds number is computed directly from $\bar{U}$, D , and ν ; the pressure difference ΔP is obtained by sampling the pressure field at two fixed points upstream and downstream of the cylinder via `VectorTools::point_value`; and the Strouhal number is estimated a posteriori from the recorded $c_L(t)$ time history using the peak-detection procedure described in Section 1.8. All benchmark quantities and, where relevant, the full time history of c_D , c_L , and ΔP , are exported to CSV files at the end of the simulation for post-processing.

3 Results

This section presents the discussion and assessment of the results obtained from the implemented solvers, interpreted in the context of the underlying numerical algorithms and the solution of the Navier–Stokes equations.

3.1 θ -method

Both driver programs set

$$\theta = 1.0,$$

so in practice the generalized θ -method scheme reduces to the *backward Euler* method. This is a deliberate accuracy/stability trade-off:

- *Stability.* Backward Euler is unconditionally stable, which is valuable for the long-horizon transient runs required by the benchmark.
- *Accuracy.* Backward Euler is only first-order accurate in time, $O(\Delta t)$.

Since the convective term uses a *semi-implicit* linearization rather than a fully consistent Newton or Picard iteration, a stability restriction on Δt should be respected to keep the error small:

$$\Delta t \leq C \frac{h}{\max_{\mathbf{x} \in \Omega} |\mathbf{u}^n(\mathbf{x})|}.$$

This produces consequences for accuracy and stability too:

- *Accuracy.* The convection linearization introduces an additional splitting error of the same order as the temporal discretization itself, $O(\Delta t)$, because $\mathbf{u}_h^n = \mathbf{u}_h^{n+1} + O(\Delta t)$. Since the base time discretization scheme is already only first-order (backward Euler, $\theta = 1$), this linearization error does not degrade the nominal temporal order of the scheme.
- *Stability.* Avoiding a nonlinear (Newton) iteration removes any risk of Newton divergence at high Reynolds numbers, at the price of using only first-order-consistent transport information. This is a standard, robust choice for benchmark-type moderate-Re flows.

3.2 Preconditioner choice

To choose which preconditioner to apply, we evaluated the computational performance of two approaches suggested in [2] by monitoring the average wall-clock time per timestep throughout a 10-core simulation. This metric provides insight into how the computational cost evolves as the solution of the Navier–Stokes equations progresses.

The definition of aSIMPLE preconditioner is given in Eq. (12); the approximate Yosida preconditioner

$$P_{\text{aYosida}}^{-1} = \begin{pmatrix} \hat{F}^{-1} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -B^T \\ 0 & I \end{pmatrix} \begin{pmatrix} F & 0 \\ 0 & I \end{pmatrix} \times \begin{pmatrix} I & 0 \\ 0 & -\hat{S}^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -B & I \end{pmatrix} \begin{pmatrix} \hat{F}^{-1} & 0 \\ 0 & I \end{pmatrix} \quad (23)$$

is also derived from its exact form following the strategy described in Section 1.5. In the preconditioner inspired by the Yosida method, the Schur complement is approximated using $\tilde{S} = \Delta t B M_u^{-1} B^T$, where M_u denotes the mass matrix associated to the velocity shape

functions. To reduce the computations, in aYosida we instead replace M_u with its lumped version $M_{u,\ell}$, i.e. a diagonal matrix where $\{M_{u,\ell}\}_{ii} = \sum_j |\{M_u\}_{ij}|$.

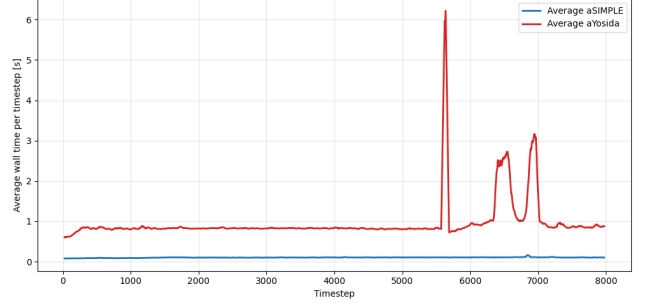


Figure 3: aSIMPLE vs. aYosida average performance across all 2D test cases (10 cores).

Figure 3 compares the average wall time per timestep for the aSIMPLE and aYosida-based implementations of the Navier–Stokes solver, across all 2D test cases. Throughout most of the simulation, the aSIMPLE preconditioning exhibits a nearly constant computational cost of approximately 0.1 s per timestep, indicating stable convergence behavior. In contrast, the aYosida implementation has a significantly higher baseline cost (around 0.8–0.9 s per timestep) and displays several pronounced spikes. The largest peak, around timestep 5600, reaches more than 6 s, while additional peaks between timesteps 6400 and 7000 increase the cost to approximately 2.5–3.2 s. However, we observed that while aSIMPLE approach generally requires 1 to 12 FGMRES iterations based on the time progression, the aYosida one achieves convergence in slightly less iterations, approximately 1 to 8 per timestep. This likely suggests that the dominant cost lies in the construction or update of the preconditioner employed by the solver. During phases where the Navier–Stokes system becomes more challenging (e.g., due to the development of strong velocity or pressure gradients, or vortex interactions) the coefficient matrices may change sufficiently to make the preconditioner more expensive to assemble. The resulting overhead can substantially increase the wall time of individual timesteps even if the number of solver iterations remains relatively low.

The isolated nature of these peaks indicates that they are associated with transient flow events rather than a sustained degradation in solver performance. Once the flow returns to a less demanding regime, the computational cost decreases to its nominal value, suggesting that the additional expense is tied to occasional updates of the linear algebra components rather than to the overall efficiency of the algorithm.

We noticed even a more significant difference when applying the two preconditioners to 3D test cases. Indeed, while the aSIMPLE approach requires around 6 s per timestep, the aYosida implementation experiences peaks where the wall time reaches 583 s. These test

runs were aborted after a low number of timesteps as the discrepancy was already clear, so no graphs about these overall 3D simulations are provided.

Given the performance analysis, we decided to proceed using just aSIMPLE preconditioner.

3.3 2D-1 test case

The steady benchmark case 2D-1 ($U_m = 0.3 \text{ m s}^{-1}$, $\nu = 10^{-3} \text{ m}^2 \text{ s}^{-1}$, giving $Re = 20$ based on the reference velocity $\bar{U} = \frac{2}{3}U(0, H/2)$) was integrated on the mesh

$$h = 0.0205,$$

with the fully implicit backward-Euler scheme using timestep

$$\Delta t = 0.001 \text{ s},$$

up to $T = 8 \text{ s}$. Although the case is nominally steady, it is still advanced in time from the zero initial condition: at $Re = 20$ the flow around the cylinder is well below the critical Reynolds number for vortex shedding, so the solution converges to a time-independent fixed point rather than to a limit cycle.

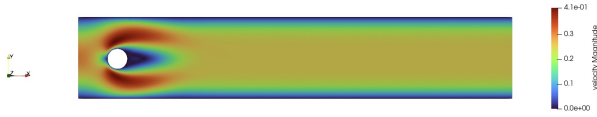


Figure 4: Velocity field at $t = 2.0 \text{ s}$ (test case 2D-1).

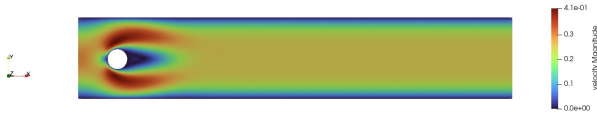


Figure 5: Velocity field at $t = 8.0 \text{ s}$ (test case 2D-1).

Figure 4 shows the velocity magnitude at $t = 2 \text{ s}$. By this point the flow has already relaxed onto the steady solution: the same field (up to differences below plotting resolution) is observed at every later sampled instant up to $t = 8 \text{ s}$, confirming that steady state has been reached well before the end of the simulation. The plot exhibits the expected qualitative features of low- Re flow past a cylinder in a channel: symmetric acceleration of the flow around the cylinder (red lobes, $|\mathbf{u}| \approx 0.41$) due to the blockage of the channel cross-section, a slow-moving symmetric wake immediately downstream of the cylinder (dark blue region), and a re-established parabolic-like profile far downstream. The symmetry of the field is itself a qualitative check on the correctness of the discretization and of the boundary treatment.

Since the flow is steady, the drag and lift coefficients $c_D(t)$, $c_L(t)$ do not carry any additional information once the transient has died out: a time series plot would simply show a flat line after the initial settling phase. So, unlike the unsteady cases 2D-

2/2D-3, where the time history of $c_L(t)$ is the object of interest, reporting the single converged value at the final time $t = T$ is both necessary and sufficient here. The table below reports the values computed by `compute_benchmark_result()` at $t = 8 \text{ s}$:

Re	c_D	c_L	L_a [m]	ΔP [Pa]
20	5.47236	0.00122345	0.08437	0.11784

These values are in good qualitative and quantitative agreement with the reference intervals of [1] for this case: the recirculation length L_a , the pressure drop ΔP , and c_L fall within the published interval, while c_D is slightly under-predicted. The fact that c_L is near but not exact zero could be due to its sensitivity to any asymmetry introduced by the mesh or residual truncation error at $t = 8 \text{ s}$.

3.4 2D-2 test case

Case 2D-2 ($U_m = 1.5 \text{ m s}^{-1}$, $\nu = 10^{-3} \text{ m}^2 \text{ s}^{-1}$, $Re = 100$) was solved with the same discretization parameters as 2D-1 ($h = 0.0205$, $\theta = 1$) and the same timestep

$$\Delta t = 0.001 \text{ s},$$

up to $T = 8 \text{ s}$. Unlike 2D-1, here $Re = 100$ is well above the critical Reynolds number for the onset of vortex shedding behind a confined cylinder ($Re_{cr} \approx 46$), so the long-time solution is not a fixed point but a time-periodic limit cycle. Using the same Δt as in the steady case is deliberate: since backward Euler is unconditionally stable regardless of the flow regime, no re-tuning of the time step is required; $\Delta t = 0.001 \text{ s}$ remains small enough to keep the $O(\Delta t)$ time-discretization error controlled relative to the shedding period.



Figure 6: Velocity field at $t = 0.5 \text{ s}$ (test case 2D-2).

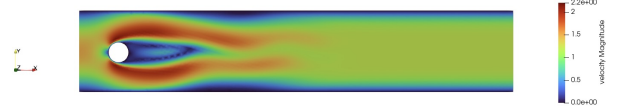


Figure 7: Velocity field at $t = 1.0 \text{ s}$ (test case 2D-2).

Figures 6–9 show the velocity magnitude at four instants, $t = 0.5, 1.0, 5.0, 8.0 \text{ s}$, illustrating how the flow evolves from the impulsive start to the fully developed periodic regime. The qualitative progression corresponds to what is expected physically: at $t = 0.5\text{--}1 \text{ s}$ the wake is still symmetric (the initial condition and boundary data are symmetric about the centerline, so

any asymmetry can only grow from the amplification of numerical/round-off perturbations); by $t = 5$ s the velocity field shows a fully alternating pattern of shed vortices. Since by $t = 8$ s the pattern looks indistinguishable from the one at $t = 5$ s, we conclude that the simulation has reached its asymptotic periodic regime.

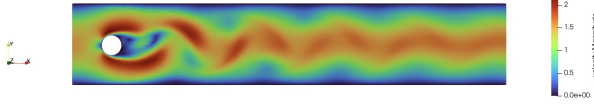


Figure 8: Velocity field at $t = 5.0$ s (test case 2D-2).

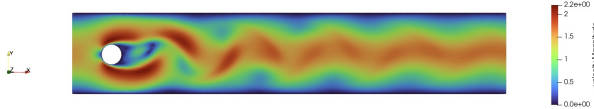


Figure 9: Velocity field at $t = 8.0$ s (test case 2D-2).

Since the asymptotic state is periodic, plotting $c_D(t)$ and $c_L(t)$ over the whole simulation window reproduces, cycle after cycle, the same oscillation already visible in the figures above (check Figure 15).

The following table reports the benchmark values actually computed for this run:

Re	$c_{D,\max}$	$c_{L,\max}$	St	ΔP [Pa]
100	3.17191	1.0456	0.300601	2.50013

These results agree well with the reference intervals of [1] for this case. Again, $c_{D,\max}$ is marginally under-predicted, consistent with the same sources of error discussed for case 2D-1.

3.5 2D-3 test case

Case 2D-3 uses the same geometry, viscosity, and spatial discretization as cases 2D-1/2D-2, but with a sinusoidally time-varying inflow amplitude,

$$U_m(t) = 1.5 \sin\left(\frac{\pi t}{8}\right) \text{ m s}^{-1}, \quad t \in [0, 8] \text{ s}.$$

As in cases 2D-1 and 2D-2, the problem was integrated with backward Euler ($\theta = 1$) and timestep

$$\Delta t = 0.001 \text{ s}.$$

The choice of the same Δt across all three cases is again consistent with the unconditional stability of backward Euler scheme.

Figures 10–14 show the velocity magnitude at five instants spanning the full ramp, $t = 0.5, 2.0, 4.0, 6.0, 7.5$ s. At $t = 0.5$ s, the imposed velocity is still very small ($U_m(0.5) \approx 0.29 \text{ m s}^{-1}$) and the whole domain is near rest. At $t = 2$ s, the flow around the cylinder resembles a quasi-steady

symmetric wake, similar in character to case 2D-1. Finally, at $t = 4$ s the inflow reaches its maximum ($U_m = 1.5 \text{ m s}^{-1}$, $Re = 100$) and the wake shows the first signs of the sinuous instability, but only weakly developed compared to the fully saturated vortex street of case 2D-2 at the same instantaneous Re .



Figure 10: Velocity field at $t = 0.5$ s (test case 2D-3).



Figure 11: Velocity field at $t = 2.0$ s (test case 2D-3).

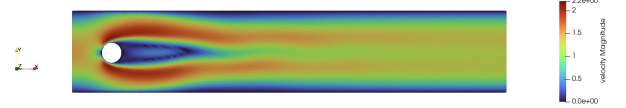


Figure 12: Velocity field at $t = 4.0$ s (test case 2D-3).

During the decaying phase of the inflow ramp, at $t = 6$ s, the wake shows a clear vortex pattern extending far downstream, in marked contrast with the still-symmetric wake observed at $t = 2$ s for the same inflow amplitude. At $t = 7.5$ s the local inflow has nearly vanished and the overall velocity magnitude is uniformly low, yet a slightly wavy structure persists in the wake, even though the driving inflow can no longer sustain it.

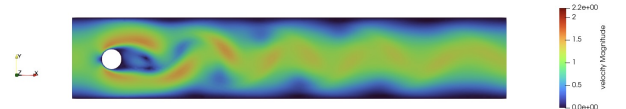


Figure 13: Velocity field at $t = 6.0$ s (test case 2D-3).

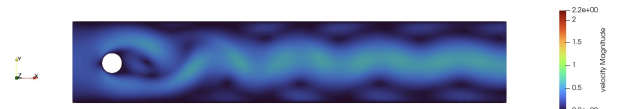


Figure 14: Velocity field at $t = 7.5$ s (test case 2D-3).

The asymmetry between the rising and the decaying phase at matching instantaneous inflow amplitudes is the key qualitative feature of this case. Indeed, in the early phase, the instability has not yet had time to develop, so the wake remains close to the quasi-steady, symmetric state expected from the instantaneous Re

alone. By the time the inflow starts decreasing, the vortices generated during the high-Re window ($t \approx 3\text{--}5\text{ s}$) are still convecting through the domain, so the wake at $t = 6\text{ s}$ and even at $t = 7.5\text{ s}$ retains a visible, decaying wavy signature (memory effect of the earlier instability), even though the local driving velocity has already got back to a sub-critical level.

Here, only the peak coefficients over the recorded history and the pressure difference at the final time are reported:

$c_{D,\max}$	$c_{L,\max}$	$\Delta P(t = 8\text{ s})$ [Pa]
2.88699	0.456861	-0.103745

Both peak coefficients are markedly lower than their 2D-2 counterparts ($c_{D,\max} = 2.887$ vs. 3.172 , $c_{L,\max} = 0.457$ vs. 1.046), which is the direct quantitative counterpart of the qualitative observation above: since the wake instability in 2D-3 never fully saturates before the forcing subsides, the lift amplitude in particular remains well below the saturated value reached in the constant-inflow case.

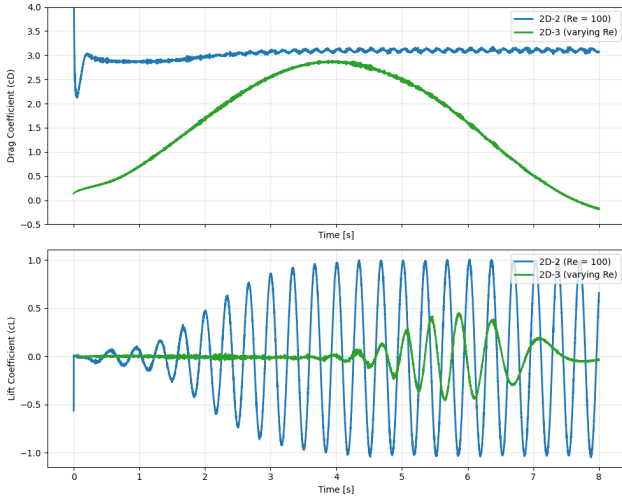


Figure 15: Drag and lift coefficient trends.

Figure 15 superimposes the drag and lift coefficient histories of the constant-inflow case 2D-2 and the time-varying-inflow case 2D-3, where the instantaneous Reynolds number ramps smoothly from $Re = 0$ at $t = 0$ to $Re = 100$ at $t = 4\text{ s}$ and back down to $Re = 0$ at $t = 8\text{ s}$. Both runs use the same spatial discretization and the same timestep $\Delta t = 0.001\text{ s}$, so the differences observed below are entirely attributable to the different inflow forcing, not to the numerical scheme.

- *Drag coefficient.* In case 2D-2, $c_D(t)$ shows a sharp, impulsive start before settling, by $t \approx 1\text{--}1.5\text{ s}$, into a small-amplitude periodic oscillation about a mean value $c_D \approx 3.0$, consistent with the fully developed vortex-shedding regime already

identified in Fig. 8. In case 2D-3, by contrast, $c_D(t)$ has no such transient spike, since the inflow itself starts from rest and grows smoothly as $\sin(\pi t/8)$. The drag coefficient peaks around $c_D \approx 2.85$ near $t \approx 4\text{ s}$, precisely when the instantaneous Re is at its maximum ($Re = 100$, matching the constant-inflow case), before decreasing symmetrically back toward zero as the inflow decays.

- *Lift coefficient.* In case 2D-2, the lift oscillation grows from an initially small amplitude to a fully saturated ± 1.0 swing by $t \approx 4\text{--}5\text{ s}$. Instead, in case 2D-3, $c_L(t)$ stays essentially at zero for the first $\approx 2\text{ s}$, and only develops a visible, low-frequency, low-amplitude oscillation (peak $|c_L| \lesssim 0.4$) in the window where the instantaneous Re is close to its maximum, $t \approx 3\text{--}6\text{ s}$. The lift oscillation for 2D-3 is thus a transient, partially developed instability rather than a sustained limit cycle.

3.6 3D-1Z test case

Case 3D-1Z ($U_m = 0.45\text{ m s}^{-1}$, $\nu = 10^{-3}\text{ m}^2\text{ s}^{-1}$, giving $Re = 20$ based on the 3D reference velocity $\bar{U} = \frac{4}{9}U(0, H/2, W/2)$) was solved on the mesh having size

$$h = 0.0410,$$

with backward Euler ($\theta = 1$) and timestep

$$\Delta t = 0.01\text{ s}.$$

This is coarser in time than all the 2D runs of Sections 3.3–3.5 ($\Delta t = 0.001\text{ s}$). As discussed previously, this larger Δt does not threaten stability as backward Euler remains unconditionally stable regardless of it, but it does increase the $O(\Delta t)$ time-discretization error of the linearized scheme during the initial transient. Since 3D-1Z is a steady case, no vortex shedding is expected, and thus this larger transient error is faded once the solution has converged to its steady state, not affecting the quality of the final, time-independent result. This choice has been made due to execution time, as 3D simulations take hours to complete.

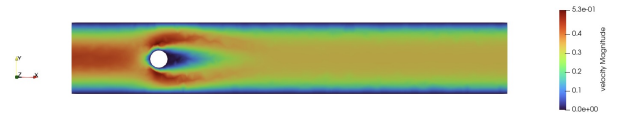


Figure 16: Velocity field at $t = 2.0\text{ s}$ (test case 3D-1Z).

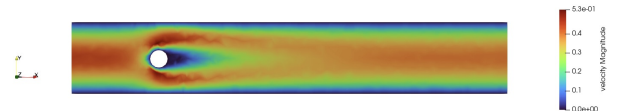


Figure 17: Velocity field at $t = 8.0\text{ s}$ (test case 3D-1Z).

Figures 16 and 17 show the velocity magnitude on the $z = W/2$ plane, at $t = 2$ s and $t = 8$ s. The two snapshots are almost indistinguishable, confirming (as in case 2D-1) that the flow has already relaxed onto its steady state well before $t = 2$ s, and that the coarser $\Delta t = 0.01$ s was sufficient to resolve that transient without leaving any visible residual unsteadiness at later times. The field exhibits the expected qualitative features of confined low- Re flow past a 3D cylinder: symmetric acceleration around the obstacle (red lobes, $|\mathbf{u}| \lesssim 0.53 \text{ m s}^{-1}$, consistent with the bi-parabolic inflow profile and the blockage of the channel cross-section), and a slow, symmetric, non-shedding wake downstream. This mirrors the 2D-1 picture of Figures 4 and 5.

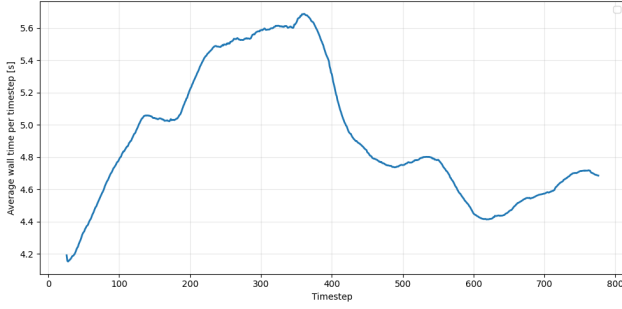


Figure 18: Rolling mean ($w=50$) wall time per timestep for 3D-1Z test case using aSIMPLE preconditioner (10 cores).

As for case 2D-1, the steady nature of this flow means the drag and lift coefficient histories carry no information beyond the initial settling transient, so the single converged value at $t = T = 8$ s is reported rather than a time series. The table below lists the values computed by `compute_benchmark_result()`:

Re	c_D	c_L	ΔP [Pa]
20	5.96773	-0.00533984	0.177818

These values are in line with the published 3D reference [1] intervals for this case. Small deviations are plausibly attributable to the combination of the coarser timestep and the affine (non-curved) approximation of the cylinder at this mesh resolution.

As in the 2D-1 case, c_L is the most sensitive quantity: at this magnitude, the lift coefficient is essentially a measure of the residual asymmetry introduced by the mesh and linear-solver tolerance rather than a physically significant result.

3.7 3D-2Z test case

Case 3D-2Z ($U_m = 2.25 \text{ m s}^{-1}$, $\nu = 10^{-3} \text{ m}^2 \text{ s}^{-1}$, $Re = 100$) was run on the same mesh and time step discussed for case 3D-1Z,

$$h = 0.0410, \quad \Delta t = 0.01 \text{ s},$$

up to $T = 8$ s, with backward Euler method ($\theta = 1$). Unlike 3D-1Z, this case is above the shedding threshold, so an unsteady, periodic-in-time regime is expected in the long run.



Figure 19: Velocity field at $t = 0.5$ s (test case 3D-2Z).

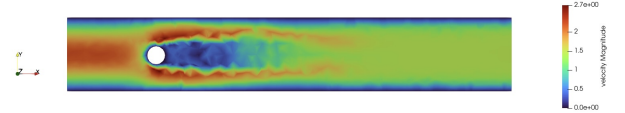


Figure 20: Velocity field at $t = 1.0$ s (test case 3D-2Z).

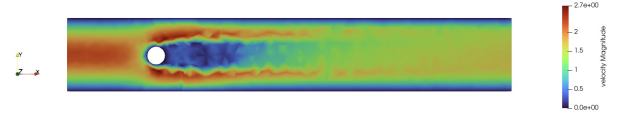


Figure 21: Velocity field at $t = 5.0$ s (test case 3D-2Z).

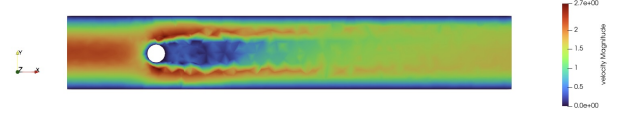


Figure 22: Velocity field at $t = 8.0$ s (test case 3D-2Z).

The figures above show the velocity magnitude on the mid-plane $z = W/2$ at $t = 0.5, 1.0, 5.0, 8.0$ s. The transient at $t = 0.5$ s (Figure 19) is still comparatively smooth and close to symmetric, similarly to the early transient of case 2D-2 (Figure 6). By $t = 1$ s, however, the wake already displays a visibly fine-grained, speckled structure in the transition region between the low-speed core of the wake (dark blue) and the recovering far wake (green), and this pattern persists essentially unchanged through $t = 5$ s and $t = 8$ s.

This is qualitatively different from the clean, large-scale, coherent fluid flow seen in the 2D case at the same nominal $Re = 100$ (Figure 9). Two distinct explanations are plausible. Physically, a three-dimensional wake at $Re = 100$ is expected to develop secondary instabilities superimposed on the primary shedding, which have no counterpart in a strictly 2D simulation and can produce exactly this kind of fine-grained, less coherent structure on a mid-plane slice. Numerically, however, the coarser discretization used here ($h = 0.0410$, $\Delta t = 0.01$ s) is also a contributor, manifesting noisy features rather than smooth, well-defined vortices. Distinguishing a genuine 3D secondary instability from an under-resolution artifact would re-

quire repeating this case at finer h and Δt and checking whether the speckled pattern persists or disappears.

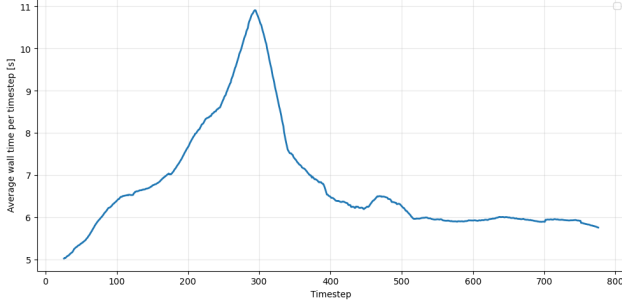


Figure 23: Rolling mean ($w=50$) wall time per timestep for 3D-2Z test case using aSIMPLE preconditioner (10 cores).

As in case 2D-2, the periodic (or quasi-periodic) nature of this flow means the peak coefficients and the shedding frequency are the relevant summary quantities rather than the full $c_D(t)$, $c_L(t)$ traces. The table reports the values computed by `compute_benchmark_result()`:

Re	$c_{D,\max}$	$c_{L,\max}$	St
100	2.97964	0.0799768	0.589744

Both peak coefficients are substantially lower than their 2D-2 counterparts ($c_{D,\max} = 2.98$ vs. 3.17, and especially $c_{L,\max} = 0.080$ vs. 1.046), which is consistent with the physical expectation that a 3D wake redistributes part of the unsteady loading into secondary instability motion, and with the more diffuse shedding pattern. The reported Strouhal number, $St \approx 0.59$, is nearly double the value found for the analogous 2D case ($St = 0.30$) and is higher than typically reported 3D benchmark values for this configuration. Given the coarse $\Delta t = 0.01$ s used here, this is a plausible symptom of the peak-picking St estimator (Section 1.8) locking onto spurious secondary maxima in a noisy $c_L(t)$ signal.

3.8 3D-3Z test case

Case 3D-3Z combines the time-varying inflow of case 2D-3 with the 3D geometry of cases 3D-1Z/3D-2Z:

$$U_m(t) = 2.25 \sin\left(\frac{\pi t}{8}\right) \text{ m s}^{-1}, \quad t \in [0, 8] \text{ s},$$

so that the instantaneous Reynolds number again ramps from $Re = 0$ at $t = 0$ up to $Re = 100$ at $t = 4$ s and back to $Re = 0$ at $t = 8$ s. As for 3D-1Z and 3D-2Z, the run uses

$$h = 0.0410, \quad \Delta t = 0.01 \text{ s}, \quad \theta = 1.$$

Figures 24–28 show the velocity magnitude on the mid-plane $z = W/2$ at $t = 0.5, 2.0, 4.0, 6.0, 7.5$ s.



Figure 24: Velocity field at $t = 0.5$ s (test case 3D-3Z).

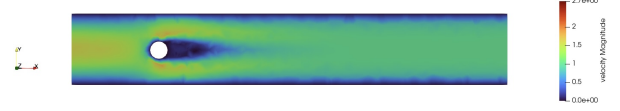


Figure 25: Velocity field at $t = 2.0$ s (test case 3D-3Z).

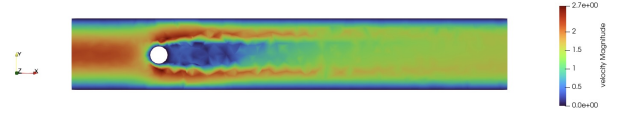


Figure 26: Velocity field at $t = 4.0$ s (test case 3D-3Z).

During the rising phase, at $t = 0.5$ s, the whole domain is close to rest. At $t = 2$ s the wake is still smooth and essentially symmetric, closely resembling the steady 3D-1Z field of Figure 16. Finally, at $t = 4$ s, the instant of peak inflow ($Re = 100$), the wake begins to show the first signs of the speckled structure identified for case 3D-2Z, but only weakly.

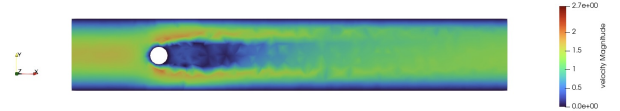


Figure 27: Velocity field at $t = 6.0$ s (test case 3D-3Z).

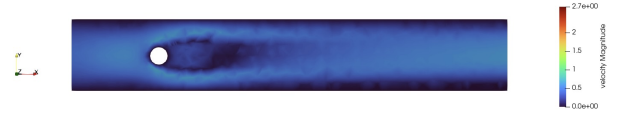


Figure 28: Velocity field at $t = 7.5$ s (test case 3D-3Z).

During the decaying phase, at $t = 6$ s (same instantaneous amplitude as $t = 2$ s), the wake shows a markedly more disturbed structure than its rising-phase counterpart, extending further downstream. At $t = 7.5$ s, with the inflow nearly back to zero, a faint speckled pattern is still discernible close to the cylinder even though the driving velocity can no longer sustain it. The asymmetry already identified for the 2D counterpart is reproduced here, but now superimposed on speckled wake pattern that distinguishes the 3D shedding cases at this level of discretization.

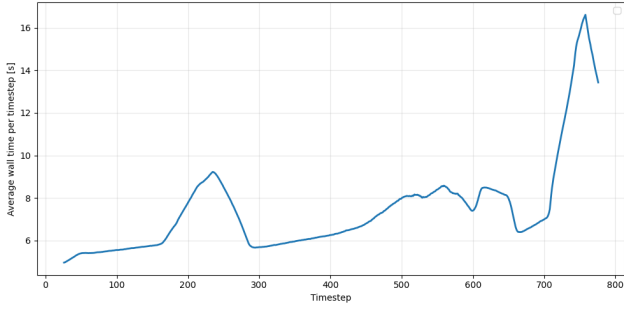


Figure 29: Rolling mean ($w=50$) wall time per timestep for 3D-3Z test case using aSIMPLE preconditioner (10 cores).

As for case 2D-3, the flow here never reaches a periodic regime, so `compute_benchmark_result()` reports only the peak coefficients over the recorded history and the pressure difference at the final time, without a Strouhal number:

$c_{D,\max}$	$c_{L,\max}$	$\Delta P(t = 8\text{ s})$ [Pa]
2.9739	0.0746074	-0.115357

The peak coefficients are close to those of the constant-inflow case 3D-2Z ($c_{D,\max} = 2.974$ vs. 2.980, $c_{L,\max} = 0.0746$ vs. 0.0800), and much closer, proportionally, than the corresponding 2D comparison (2D-3 vs. 2D-2), where $c_{L,\max}$ dropped to well under half its constant-inflow value. The final-time pressure difference is again small and slightly negative (-0.115 Pa), consistent with the inflow having nearly returned to zero and the residual convected wake structure of Figure 28 still relaxing. As with case 3D-2Z, any quantitative conclusion drawn from these peak values should be read together with the caveat that $h = 0.0410$, $\Delta t = 0.01$ s is a relatively coarse discretization for capturing the fine-scale 3D wake structure, and a resolution check would be needed before treating these numbers as fully converged.

Figure 30 compares the drag and lift histories of the constant-inflow case 3D-2Z (blue) and the time-varying-inflow case 3D-3Z (green). Both runs share the same spatial and temporal discretization, $h = 0.0410$, $\Delta t = 0.01$ s.

- *Drag coefficient.* The graph mirrors the 2D comparison exactly: 3D-2Z shows a sharp initial transient before settling, by $t \approx 1$ s, onto an essentially constant plateau at $c_D \approx 3.0$. Case 3D-3Z instead shows no such impulsive transient: $c_D(t)$ rises smoothly, and peaks close to the 3D-2Z plateau value ($c_D \approx 3.0$) right around $t \approx 4$ s, the instant of maximum inflow, before decreasing symmetrically as $U_m(t)$ decays.
- *Lift coefficient.* While $c_L(t)$ for 2D-2 saturates at a fast ± 1.0 oscillation (Figure 15), and 2D-3's lift

never approaches that saturated amplitude, in 3D both curves are small and slowly varying. 3D-2Z's $c_L(t)$ settles into a shallow, low-frequency oscillation around $c_L \approx -0.08$, while 3D-3Z's $c_L(t)$ traces a smooth dip down to about -0.075 near $t \approx 3$ –4 s and back up toward zero as the inflow decays. In other words, the 2D pattern does not repeat nearly as strongly in 3D: since the 3D wake's coherent-shedding contribution to c_L is already comparatively small and diffuse (Section 3.7), 3D-3Z's lift history tracks the constant-inflow case fairly closely in both sign and magnitude.

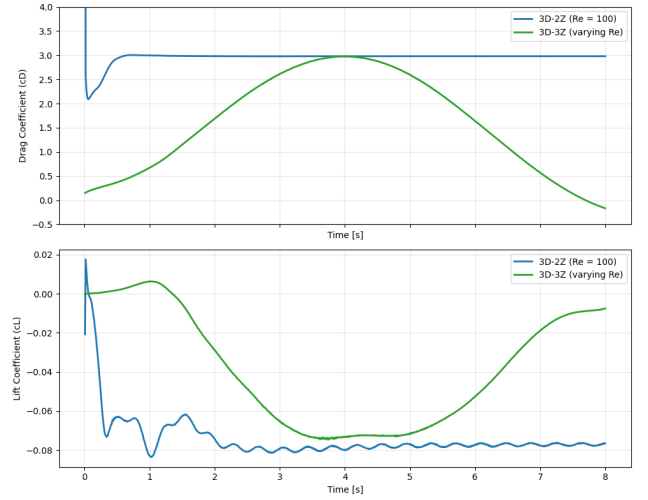


Figure 30: Drag and lift coefficients trends.

This alignment between the two 3D cases is exactly what was already noted when comparing their respective coefficient tables: $c_{D,\max}$ and $c_{L,\max}$ for 3D-3Z sit much closer, proportionally, to the 3D-2Z values than 2D-3's did to 2D-2's.

References

- [1] M. Schäfer and S. Turek (with F. Durst, E. Krause, R. Rannacher), *Benchmark Computations of Laminar Flow Around a Cylinder*, in: E.H. Hirschel (ed.), *Flow Simulation with High-Performance Computers II*, Notes on Numerical Fluid Mechanics, Vol. 52, Vieweg, 1996, pp. 547–566.
- [2] S. Deparis, G. Grandperrin, and A. Quarteroni, *Parallel preconditioners for the unsteady Navier–Stokes equations and applications to hemodynamics simulations*, *Computers & Fluids*, 92 (2014), pp. 253–273.